Harish Chaurasia

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SUMMARY

AI Engineer building production LLM systems - agentic workflows, RAG pipelines, and evaluation frameworks - across TypeScript, Python, FastAPI, and cloud infrastructure. Hands-on with LLM fine-tuning (LoRA/QLoRA), agent reliability, and shipping full-stack AI features end-to-end.

EDUCATION

Arizona State University, Tempe, AZ

Master's in Computer Science (AI/ML) GPA: 3.78 | May 2026
Relevant Courses: Agentic AI, Robotics Learning, Natural Language Processing, Cloud Computing, AI/ML
Bachelor's in Computer Science (**Minor:** Game Development) GPA: 3.59 | May 2025
Relevant Courses: Data Structures & Algorithms, Software Engineering, DBMS, Artificial Intelligence

EXPERIENCE

AI Engineer (Jr. SWE), Built By The Trades (BBTT), Full Time **Aug 2026 - Oct 2026**

- Built and maintained **Foreman**, a Claude-API help assistant grounded on internal guides with safety fallbacks and out-of-scope refusal.
- Owned **ServiceDash** end-to-end - a real-time home-service marketplace spanning homeowner, contractor, technician, and admin apps - from database and backend through to the UI.
- Drove UI/UX work across four apps on a new design system with light/dark themes, improving live tracking, scheduling, and ratings flows.

Software Developer, MSN School, Ira A. Fulton Schools of Engineering **Dec 2023 - May 2026**

- Embedded with Los Alamos National Laboratory technical stakeholders to scope, design, and deploy high-fidelity VR training modules; ran iterative discovery sessions and shipped to end-users, improving immersive training accuracy by **30%**.
- Owned the full delivery lifecycle - discovery, prototype, iteration on user feedback, and production rollout - navigating ambiguous requirements and shifting operational needs from a regulated national-lab customer.

Software Engineer, Laboratory for Energy and Power Solutions (LEAPS) **Jan 2023 - Oct 2023**

- Delivered AR/VR microgrid training systems to **10+** institutional partners as sole engineer on the deployment side; handled scoping, integration, and post-deployment adoption support directly with partner sites.

PROJECTS

WebSecArena - Agent Reliability Evaluation Framework *(Python, BrowserGym, AgentLab)*

- Designed and built an end-to-end evaluation framework for LLM web agents across 6 attack classes (prompt injection, phishing, clickjacking, etc.); ran **2,160** trials across 12 agent-model combinations to quantify production reliability before deployment.
- Codified a 3-outcome scoring methodology (ASR, STS, DTR) that separates genuine agent robustness from overcaution - a reusable evaluation pattern surfacing **60-100%** baseline failure rates on realistic attack vectors.

Compass - Training-Free Confidence Calibration for Production Agents *(Python, ReAct, LLM APIs, MCP)*

- Built a training-free wrapper that gates every agent action - execute, self-verify, or abstain - using Bayesian confidence discounting and oscillation-based decay; deployable on any frontier or local LLM with no fine-tuning.
- Validated on live config restoration with Llama 3.1 8B: vanilla ReAct silently overwrote the target file and reported success; Compass abstained at **0.40** calibrated confidence, preventing destructive action in production-grade conditions.

AutoResearch - Multi-Agent Pipeline with Production Cost Discipline *(Python, OpenAI API, Pydantic)*

- Architected a 5-agent hierarchical pipeline (Planner, Retriever, Summarizer, Critic, Writer) that decomposes research questions and produces structured Markdown outputs via iterative self-critique.
- Enforced all inter-agent messages via validated Pydantic schemas with full run tracing and per-agent model configuration; observed cost of **$0.01-$0.03** per run for predictable production economics.

Codebase-MCP - Production Agent Artifact for Developer Workflows *(Python, MCP, Pydantic, AST)*

- Built and shipped a MCP server that drops directly into Claude Desktop and Cursor as a production agent tool, letting AI assistants reason about real customer repositories via AST parsing, dependency graphs, TF-IDF search, and persistent memory.
- Implemented a rule-based agent loop with tool registry, planning, and self-evaluation; designed as a reusable integration pattern for enterprise developer environments with minimal configuration.

Knowledge Brain - Privacy-Preserving RAG Deployment *(FastAPI, Next.js, FAISS, PyTorch, Llama 3, Whisper)*

- Built a fully local, VPC-friendly RAG system ingesting PDFs, documents, and audio via chunked embeddings, FAISS retrieval, and grounded Llama 3 inference; designed for customer document workflows with strict data-residency constraints.
- Optimized retrieval quality using hybrid dense-sparse search with re-ranking; kept latency under **2 seconds** for multi-document queries while maintaining full local deployment for enterprise privacy requirements.

LEADERSHIP

Technical Officer & Treasurer, Google Developer Student Club, ASU **Jan 2022 - May 2026**

Enabled **750+** students through hands-on workshops on AI/ML, cloud, MLOps, and full-stack development; authored technical playbooks and organized hackathons and bootcamps used as reusable onboarding material across cohorts.

TECHNICAL SKILLS

Languages & Tools: Python, C++, Java, C#, JavaScript/TypeScript, Swift, SQL, PostgreSQL, Claude Code, Codex, Cursor
AI/ML: PyTorch, TensorFlow, Hugging Face, Scikit-learn, RAG Pipelines, MCP, AI Agents, LLM Finetuning, AI Safety,
Cloud/DevOps: Azure, AWS, Docker, Kubernetes, Kafka, CI/CD, Git, GitLab, Datadog, Linux/UNIX
Frameworks: Node.js, FastAPI, React, Django, Langchain/Langraph, .NET Core, vLLMs, ollama